Modified TLS protocol using BB84 protocol and Bit Masking

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Abstract. This paper proposes Quantum Layer Security (QLS) to modify the existing TLS protocol, integrating the BB84 quantum key distribution for enhanced security. We propose a novel extension to the BB84 protocol that improves the secret key bit-level security by leveraging bit masking, making interception and brute force difficult. We also simulated and benchmarked the quantum system using IBM Qiskit and measured the security and performance trade-offs. The QLS framework presents an upcoming solution and a transition for securing communications in the quantum era, bridging classical and quantum cryptographic protocols, and ensuring data security against evolving threats.

Keywords: Network Security · Quantum Key Distribution · Bit-Masking

1 Introduction

1.1 TLS Protocol

Transport Layer Security (TLS) is a network security protocol that implements secure communication over a computer network. It provides end-to-end security, which ensures that data sent between 2 endpoints remains confidential and untampered. TLS has become the de-facto standard for privacy encryption and is being utilized in numerous applications such as web browsing, email communication, and online banking, underscoring its importance in preserving the security and privacy of Internet communications. [1]

TLS has undergone several iterations since its inception, evolving from its predecessor, the Secure Sockets Layer (SSL)—the evolution from SSL to TLS aimed to address security vulnerabilities and improve protocol efficiency. TLS 1.0, introduced in 1999, offered enhanced security over SSL 3.0. TLS 1.1 added more protection for CBC attacks while also fixing other bugs [2], and TLS 1.2 (2008) was modified to improve flexibility in the negotiations of various cryptographic algorithms [3].

The most recent version, TLS 1.3, was finalized in August 2018 and represents a significant leap forward in security and performance. TLS 1.3 focuses more on modularity than back compatibility, allowing for more straightforward implementation and less reliance on outdated and insecure cryptographic algorithms. By incorporating state-of-the-art encryption methods while removing redundant

algorithms, TLS 1.3 addresses many vulnerabilities in earlier versions, making it the most secure version of TLS. [4]

In this work, we shall focus on the key distribution aspect of TLS 1.3 and its shortcomings, which we propose to address by proposing a modification to the algorithm.

1.2 Key Distribution

Key Distribution occurs during the handshake process, during which cryptographic keys are exchanged to establish a secure and private session.

In TLS (Transport Layer Security), key distribution ensures secure communication by enabling the client and server to exchange cryptographic keys securely. During the TLS handshake, asymmetric encryption (typically RSA or Diffie-Hellman) is used to exchange a pre-master secret, which both parties use to derive the session keys. These symmetric session keys encrypt and decrypt the data exchanged during the session. This hybrid approach combines the efficiency of symmetric encryption with the security of asymmetric key exchange, ensuring that the communication remains confidential even if the session is intercepted without the keys. [5]

TLS 1.3 also significantly reduces the number of round trips required to establish a secure connection. This was done by reducing the number of potential algorithms in the cypher suite, allowing for reduced latency. TLS 1.3 also supports Zero Round-Trip Time (0-RTT) resumption, which provides near-instantaneous session resumption for returning users [5]. This is achieved by utilizing pre-shared keys (PSKs), allowing the transmission of early data to resume connection if the early data is still valid.

By incorporating these enhancements, TLS 1.3's key distribution protocol significantly improves security and efficiency, marking a considerable advancement over previous protocol versions.

1.3 Challenges

The development of quantum computers represents a major leap in computational power, enabling the processing of complex problems at unprecedented speeds. Quantum computers leverage quantum bits (qubits) to perform operations that classical computers struggle with, making them a potential threat to existing cryptographic systems.

They present significant challenges to TLS 1.3 key distribution due to their potential to undermine classical cryptographic methods. Shor's algorithm [6], for example, can efficiently factor large integers and solve discrete logarithms, which directly threatens public-key cryptography methods such as RSA [7] and elliptic curve cryptography (ECDSA, ECDHE) used in TLS 1.3. This vulnerability jeopardizes the security of key exchanges and, consequently, the confidentiality of communications. As quantum computers advance, there is an urgent need to develop and adopt specific algorithms to ensure the robustness of key distribution mechanisms against quantum-based threats.

Post-quantum cryptographic algorithms like Falcon and Dilithium provide security against quantum computing threats, ensuring safer communication. However, they remain susceptible to side-channel attacks [8], where attackers exploit physical cues like power consumption, timing, or electromagnetic emissions to extract sensitive information. These attacks bypass the cryptographic strength of the algorithms, targeting the implementation itself. Therefore, while quantum-resistant algorithms enhance security, addressing side-channel vulnerabilities is essential to achieving a truly secure, future-proof communication system.

The rest of the paper is organized as follows: Section 2 discusses some existing work in network security and quantum algorithms, Section 3 focuses on the proposed modifications, Section 4 on the experimental setup and corresponding result and finally, section 5 deals with future works and conclusion.

2 Literature Review

Sikeridis et al. [9] compared the performances of TLS when using different PQ algorithms. This was done using OQS OpenSSL to determine the latency and duration of various algorithms in other applications. The authors were able to conclude the best asymmetric algorithm for authentication in different scenarios.

Elboukhari et al. [10] proposed a novel implementation of TLS that uses BB84 for Key Distribution, which is used to share a pre-master secret. This secret is subsequently used to create a master secret for mutual authentication. The author finally worked on a feasibility study in which they talked about the physical implementation of this setup.

Adu-Kyere et al. [11] implemented a proof of concept of BB84 using IBM's QISKIT simulator and tested the program in different real-world scenarios. Various operations such as base reconciliation, rectification, and error correction were done to test the simulation and check the consistency with existing literature.

Neha et al. [12] implemented a modified version of BB84 using IBM's QISKIT simulator. The authors used the shared bits to generate a key using polynomial interpolation instead of using the shared bits directly, which helps in adding further security to the algorithm.

Crockett et al. [13] looked at various TLS 1.2 and 1.3 implementations with Post Quantum Algorithms. The authors benchmarked OpenSSL, OpenSSH and Amazon's s2n to investigate any limitations or flaws in the design of the products.

Singh et al . [14] created a survey of various QKD protocols, and it is clear that many protocols acted as minor variations to the original BB84 protocol. BB84 acts as the standard for all QKD protocols, and other protocols don't modify the key directly, so by updating the original BB84 protocol, we can extend it to the rest

3 Proposed Methodology

We propose a methodology that modifies existing TLS by replacing traditional asymmetric cryptographic systems used in key distribution with a modified ver-

sion of the BB84 protocol, addressing emerging security challenges and enhancing system resilience. The BB84 protocol, based on quantum mechanics, is inherently secure due to the detectable disturbances caused by eavesdropping [15].

However, we refine the protocol by applying an XOR operation to groups of K bits in the secret key, which obfuscates the relationship between revealed and unrevealed bits, thereby improving security without discarding the revealed bits, a limitation of the original protocol. This modification not only strengthens the system against potential future vulnerabilities but also allows for more efficient key retention.

Larger values of K enhance security by increasing the difficulty for eavesdroppers to compromise the key, though this comes with higher computational and transmission demands. To balance security with system efficiency, the value of K should align with the memory architecture, ideally being a multiple of half-word or word sizes, ensuring the system's performance remains optimized while maintaining high levels of cryptographic security.

The details of the BB84 algorithm are presented in the following section, followed by the details of the proposed modification to the algorithm.

3.1 BB84 Protocol

The BB84 protocol, introduced by Charles Bennett and Gilles Brassard in 1984 [16], is a pioneering quantum key distribution (QKD) scheme that enables two parties to securely exchange cryptographic keys using quantum mechanics. The protocol is based on the principles of quantum superposition and the no-cloning theorem, which ensure that any eavesdropping attempt on the quantum communication channel introduces detectable disturbances. The following is the description of the protocol working principle:

Preparation of Quantum States: The sender (Alice) prepares a series of quantum bits (qubits) in one of four possible states, chosen randomly from two conjugate bases (rectilinear and diagonal). These states are typically represented as the polarization states of photons.

Transmission: Alice sends these qubits to the receiver (Bob) over a quantum channel. The qubits are transmitted one at a time, ensuring that each qubit is isolated during transmission.

Measurement by Receiver: Upon receiving the qubits, Bob measures them using a randomly chosen basis (rectilinear or diagonal). Since the measurement basis is selected independently of Alice's preparation basis, there is a 50% chance that Bob's measurement basis will match Alice's preparation basis.

Public Discussion: After the transmission, Alice and Bob publicly discuss the bases they used for preparation and measurement, but not the actual results. They discard the bits where their bases do not match.

Key Generation: The remaining bits, where the preparation and measurement bases match, form the raw key. Alice and Bob then perform classical post-processing steps, such as error correction and privacy amplification, to produce a final secure key.

Eavesdropping detection: Alice and Bob reveal some parts of the key obtained and verify that there have been no discrepancies in the keys distributed. In case there is some bit mismatch, an Eavesdropper Eve has been detected, and the keys are dropped immediately.

The BB84 protocol's security stems from the fact that any attempt to eavesdrop on the quantum channel would alter the qubit states, introducing detectable errors. This feature makes BB84 a robust method for secure key distribution in quantum cryptography.

The BB84 protocol defends against classical vulnerabilities like those in Diffie-Hellman, such as the Logjam attack, by using quantum mechanics for key distribution rather than mathematical functions. In BB84, keys are encoded in quantum states that cannot be measured or copied without detection. Any interception alters these states, introducing detectable errors that reveal eavesdropping attempts, thus safeguarding the key exchange from classical weaknesses.

However, BB84 remains susceptible to side-channel attacks on hardware implementations [17], where attackers may exploit power consumption, timing, or electromagnetic emissions from quantum circuits. While quantum systems share this vulnerability with classical systems, emerging defences aim to mitigate these risks. Additionally, ongoing advancements in algorithm development [18] and future developments [19] in quantum side-channel defence are essential to strengthening quantum algorithms against such side-channel exploits for secure quantum communication.

3.2 Modification

The BB84 algorithm requires the disclosure of a random subset of bits from the shared key to detect potential eavesdropping. In the standard implementation, this step is crucial for determining whether the communication has been compromised. However, the revealed bits must be discarded from the final key to maintain confidentiality. This process, while essential for security, leads to a reduction in the length of the usable secret key, potentially limiting the overall efficiency of the protocol.

We propose an alternative approach that mitigates this issue by introducing a method that enhances security without necessitating discarding revealed bits. Our method involves dividing the secret key into groups of K consecutive bits and applying an XOR (exclusive OR) operation to these groups. Specifically, the XOR operation is used for each set of K bits, resulting in a compressed bit for the new key. This transformation creates a derived key where each bit depends on a group of four bits from the original key.

The primary advantage of this approach is that it significantly improves security by making it necessary for the attacker to obtain all K bits in the correct

sequence to determine the corresponding bit in the compressed key. As a result, the probability of an eavesdropper successfully intercepting enough information to compromise the key is reduced.

Revealing certain bits of the secret key during transmission can establish a relationship with the remaining key. Even when the revealed bits are discarded, this relationship might persist, posing a potential vulnerability in the system. An eavesdropper could use this partial information to gain insights into the key structure, increasing the risk of compromising the security of the communication. [20]

To address this issue, the application of an XOR operation introduces an effective method to mask the underlying bits, as shown in Fig. 1. By combining groups of bits through XOR, the relationship between the original and revealed bits becomes much more complex. The XOR operation not only ensures that the individual bits are concealed but also makes it extremely difficult to derive any direct information about the remaining key from the revealed bits [21]. This obfuscation scrambles the underlying relationships between bits, mitigating the potential for eavesdroppers to exploit revealed information. This significantly strengthens the security of the key, as it reduces the risk of any relationship between the disclosed and undisclosed bits being leveraged to compromise the key during quantum key distribution or any other cryptographic process.

In the original BB84 protocol, revealing individual bits allows for straightforward detection of tampering, but it also creates a vulnerability, as those bits are no longer secret. In contrast, our method increases the complexity of the attack. The probability that all K consecutive bits in any sequence will be revealed is on the order of $O(N^{-K})$, where N denotes the length of the key. This probability becomes negligible for sufficiently large values of N, ensuring that the key remains secure.

The selection of the parameter K is closely tied to the system's constraints [22], as it directly impacts the size of the key being transmitted, the available computational resources, and the characteristics of the quantum communication channel. Larger values of K contribute to improved security because the number of bits required to reveal the key increases, making the key more difficult for potential eavesdroppers to decipher. However, this enhanced security comes at a cost. As K increases, the system's demands also grow due to the appropriate scaling of underlying hardware [23], particularly regarding transmission bandwidth and computational overhead. This is due to the need to handle and transmit a more significant number of bits.

Moreover, as K increases, the development of efficient quantum hardware accelerators for key generation becomes more challenging. Larger key sizes require more advanced and resource-intensive hardware, potentially limiting the feasibility of deploying such systems at scale, especially with current quantum hardware technologies. Furthermore, due to the number of qubits being transported in the system, there exist more levels of noise in the quantum circuit, which negatively impacts the circuit output [24]

To optimize system performance, it is desirable to keep K in line with the memory alignment of the system's architecture. Ideally, K should be a multiple of 4 or 8, as this facilitates more efficient processing and transmission. Proper alignment with the underlying hardware can help mitigate some of the additional resource demands caused by larger key sizes, ensuring a balance between security and system efficiency. Therefore, careful consideration of K is essential in designing a secure and scalable quantum key distribution system.

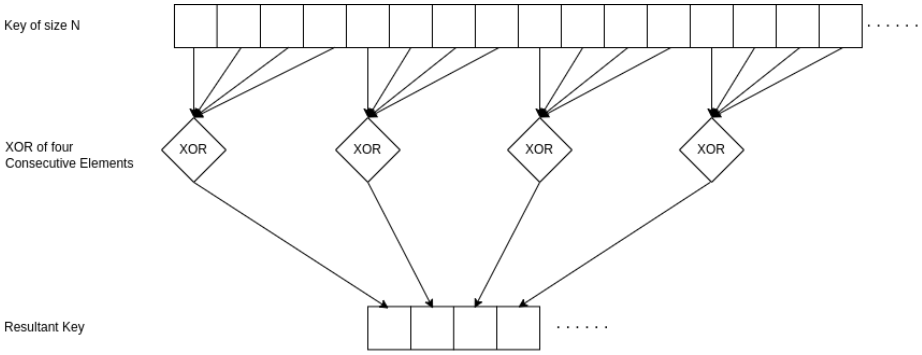


Fig. 1. XOR of K consecutive elements

4 Experimental Setup and Results

In this experiment, we simulate a quantum key distribution system using IBM Qiskit and Python's socket library to model communication between two parties. The focus is on assessing the impact of varying sizes of the parameter K in the modified BB84 protocol on system efficiency and security. K represents the number of consecutive bits combined using an XOR operation to compress the key, which affects both the security and computational requirements of the protocol. By experimenting with different sizes of K (in multiples of 4), we aim to understand how this parameter influences the trade-off between key generation speed, execution time, and overall system performance.

Execution time plays a critical role in determining the practicality of quantum key distribution in real-time systems. Shorter execution times ensure faster key generation, which is crucial for time-sensitive applications. However, increasing K can improve security by making it harder for eavesdroppers to infer the key, but it also increases the computational load. By measuring execution time for different K values, we assess how this impacts system responsiveness and the cost of the required hardware. By analyzing these metrics, we gain a comprehensive understanding of the trade-offs between security, efficiency, and computational performance under varying key sizes and security.

The experimental setup involves two parties, Alice (sender) and Bob (receiver), who obtain a shared key using the BB84 protocol. We define the key size N per the sizes of the keys used in modern symmetric cypher algorithms such as AES. Additionally, we introduce the parameter K, which corresponds to the K-folding modification. This means the total number of qubits sent by the algorithm is K×N times.

During transmission, if the key obtained at the end of the algorithm is smaller than the expected size due to misaligned polarization axes, we pad the key with zeros to ensure uniform key length. The time of execution is measured as the duration it takes for both parties to receive the final shared key, including the post-processing steps like error correction and privacy amplification. By analyzing the time of execution under different K values, we assess the trade-off between computational complexity and security, providing insights into the system's efficiency for practical applications, as given in Table 1.

Table 1. Time of execution (in milliseconds) [Key Size vs. K-Fold].

Key Size(N)	K = 4	K = 8	K = 16	K = 32	K = 64
64	1448	2732.25	5645.97	11074.12	23164.12
128	2803.97	5519.76	11021.94	22745.47	46957.29
256	9810.51	19704.95	40095.28	81893.94	179638.38
512	19194.56	39913.81	82449.789	181022.67	437145.32

The results reveal a significant increase in computational time as the parameter K grows, highlighting a clear trade-off between enhanced security and execution time. As K increases, the complexity involved in maintaining the quantum state during communication results in longer processing times. Higher K values indeed offer improved security by increasing the robustness of the quantum key distribution (QKD) process, but this enhancement comes with a higher computational demand. This growing demand poses challenges for the development of hardware tailored for QKD, as it must efficiently manage the increased workload while preserving the delicate quantum states between communicating parties. Consequently, while higher K values strengthen security, they also necessitate advanced hardware solutions capable of handling the increased computational load and maintaining the integrity of quantum states throughout the communication process.

Selecting an appropriate K value, therefore, involves balancing these security gains [25] with the practical limitations of hardware and processing efficiency. To guide this selection, a cost mechanism can be implemented to evaluate the trade-off based on the sensitivity of the data being protected and the allowable latency for communication. For high-stakes applications where confidentiality is paramount, a higher K may be warranted despite the additional computational cost. Conversely, less sensitive applications may prioritize efficiency by using a lower K value. This approach allows organizations to align their security needs with available resources, ensuring that QKD implementations remain both secure

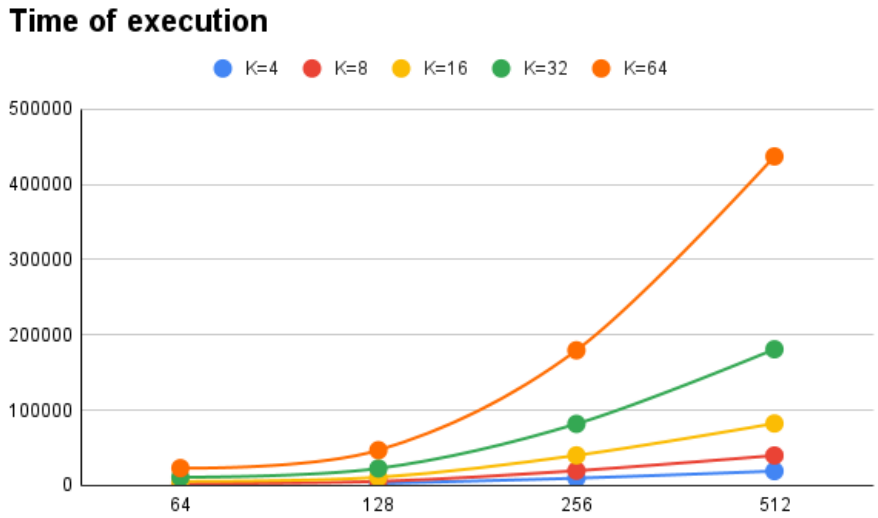


Fig. 2. Time of execution (in milliseconds) [Key Size vs. K-Fold].

and operationally viable. Advanced, adaptable hardware solutions and ongoing performance assessments will be essential to support this balance as quantum communication technology evolves.

5 Future Works & Conclusion

Quantum Key Distribution (QKD) algorithms like BB84 provide a method for secure key transmission by using quantum mechanics to detect eavesdropping. However, QKD does not address the verification of the receiver's authenticity. In TLS 1.3, this issue is resolved through the use of X.509 certificates [26], which authenticate the sender's identity using public-key cryptography methods such as RSA and elliptic curve cryptography (ECC). QKD may provide more reliable security in a post-quantum world and has models to implement the architecture, but it still faces many difficulties due to the high cost and complexity of implementation in a large-scale network [27]

Unfortunately, these traditional cryptographic algorithms are vulnerable to quantum attacks. For instance, Shor's algorithm can efficiently solve problems like integer factorization and discrete logarithms, which undermine RSA and ECC.

To counteract this, post-quantum cryptographic algorithms are being developed to resist quantum-based attacks. Algorithms like Falcon and Dilithium offer alternatives for digital signatures and public key encryption that are designed to be resistant to quantum decryption methods [28]. Integrating such post-quantum

cryptographic algorithms into TLS 1.3 could secure key distribution and certificate signing against future quantum threats, ensuring that communication remains protected as quantum computing technology evolves.

Quantum Key Distribution (QKD) protocols like BB84 are vulnerable to attacks [29] such as Photon Number Splitting (PNS), faked-state attacks, and Trojan-Horse attacks. These exploit hardware imperfections, like weak photon sources and detector flaws. Countermeasures, including decoy states and optical isolators, are being developed and are essential to mitigate these vulnerabilities and secure QKD systems.

In conclusion, the proposed Quantum Layer Security (QLS) framework integrates the BB84 quantum key distribution protocol with TLS 1.3 to address emerging security challenges posed by quantum computing. By modifying BB84 through the introduction of K-folding, the system strengthens the cryptographic security of shared keys, making them more resistant to eavesdropping and brute force attacks. Applying bitmasking to the revealed key enhances security by adding a layer of obfuscation; it makes it more challenging for potential eavesdroppers to extract the original key, even if partial information is intercepted. However, the increased computational time associated with larger K values highlights a trade-off between enhanced security and execution speed. Additionally, the inclusion of post-quantum certificates ensures robust authentication mechanisms in a quantum-resistant environment. The QLS framework bridges classical and quantum cryptographic protocols, offering a scalable and secure solution for future-proof communication systems in the quantum era.

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